

EXACT GRAM OBSTRUCTIONS AND FIVE-ROW LOCAL OPTIMALITY IN THE ORDER-23 MAXIMAL DETERMINANT PROBLEM

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ABSTRACT. The exact maximal determinant of a 23×23 sign matrix remains unknown. Let

$$L = 2,779,447,296,000,000$$

be the largest determinant currently represented by a published order-23 matrix. We prove two exact structured results at this frontier. First, no Ehlich block matrix of order 23 and determinant greater than L^2 is the Gram matrix of a sign matrix. The proof classifies all 1,255 block partitions: 894 exceed L^2 , only sixteen of these have square determinant, and every square candidate is excluded by a quadratic identity on finitely many normalized column block sums. We also exclude every sign Gram above the record within six edge toggles of the balanced Ehlich graph. Finally, a symmetric matrix attaining L cannot be improved by arbitrary simultaneous replacement of at most five rows or at most five columns. This is certified for all $\binom{23}{5} = 33,649$ five-subsets by an exact five-dimensional zonotope reduction and exterior-algebra bounds. These results exclude two substantial structured classes but do not determine the global value of $D(23)$.

1. INTRODUCTION

For a positive integer n , the Hadamard maximal determinant problem asks for

$$D(n) = \max\{|\det H| : H \in \{-1, 1\}^{n \times n}\}.$$

The exact values are presently known through order 22 [3, 4, 2]; a primary-source search through August 17, 2026 found no determination of order 23, the first unresolved case in the standard table. The largest published order-23 determinant is

$$(1) \quad L = 2^{22} 3 5^6 67 211 = 2,779,447,296,000,000,$$

represented by an explicit sign matrix [7]. Ehlich's universal bound [5, 2], together with the divisibility of every odd-order sign determinant by 2^{n-1} ,

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gives the exact integral interval

$$(2) \quad 2^{22} \cdot 662,671,875 \leq D(23) \leq 2^{22} \cdot 711,034,613.$$

This paper does not close the interval (2). Its purpose is to resolve two finite but nontrivial parts of the remaining problem by exact arguments. The first concerns the full family of relaxed Gram matrices that have Ehlich block form. Although this family supplies the sharp analytic bound, a positive definite matrix with the correct congruences need not be a sign Gram matrix. We show that every block member above (1) fails sign-Gram realizability. The key is to pass from a hypothetical factorization $G = HH^T$ to the inverse identity $H^T G^{-1} H = I$ and then sum elementary quadratic identities in the column block sums.

The remaining results quantify two different neighborhoods. On the Gram side, we classify every graph within six edge toggles of the balanced Ehlich graph, including all non-block perturbations in that radius. On the design side, we study the geometry around a particular symmetric record matrix. Replacing k rows is equivalent to maximizing a k -dimensional volume over a sign-sum zonotope. For $k = 5$, a direct enumeration of five vertices would be too large. We instead enumerate a certified superset of the vertices of the central arrangement, split the determinant through $\Lambda^2 \mathbb{R}^5$, and optimize the fifth row by an exact support function. This proves local optimality under every replacement of at most five rows. Symmetry of the record matrix yields the column statement.

All decisive computations use integer or rational arithmetic. Numerical searches are not used as proof. The exact value of $D(23)$ remains open because no known theorem reduces every high-determinant Gram matrix to either of the structured classes treated here.

Orrick reported fourteen distinct designs attaining the same order-23 record, found by gradient ascent, but did not claim a complete classification [6]. At smaller difficult orders, complete candidate-Gram generation followed by exact decomposition tests has proved successful [1]. These precedents motivate the exact Gram obstructions below while also emphasizing the missing global classification step.

2. PRELIMINARIES

For a sign matrix H of order n , its row Gram matrix $G = HH^T$ is positive definite when H is nonsingular, has diagonal n , and satisfies the familiar parity constraints on its off-diagonal entries. Moreover,

$$(3) \quad \det G = (\det H)^2.$$

Thus a necessary condition for an integral Gram candidate is that its determinant be a perfect square.

For a partition $r = (r_0, \dots, r_{s-1})$ of 23, define the Ehlich block matrix

$$(4) \quad E(r) = 20I + 4 \operatorname{diag}(J_{r_0}, \dots, J_{r_{s-1}}) - J,$$

where J_t denotes the $t \times t$ all-ones matrix. Reordering the blocks only gives a permutation-congruent matrix, so integer partitions, rather than ordered compositions, parametrize the family up to the relevant equivalence. The matrix determinant lemma gives

$$(5) \quad \det E(r) = 20^{23-s} \prod_{i=0}^{s-1} (20 + 4r_i) \left(1 - \sum_{i=0}^{s-1} \frac{r_i}{20 + 4r_i} \right).$$

We shall also use the following standard replacement identity. It is stated in a scaled form convenient for exact computation.

Lemma 2.1 (Row replacement). *Let H be nonsingular. Fix $S = \{i_1, \dots, i_k\}$ and write $q_j = H^{-1}e_{i_j}$. If the rows indexed by S are replaced, in their original order, by row vectors $x_1, \dots, x_k$, then*

$$\frac{\det H'}{\det H} = \det(x_a q_b)_{a,b=1}^k.$$

Proof. The matrix $H'H^{-1}$ agrees with the identity outside the rows in S . Laplace expansion along those unchanged rows, or equivalently the matrix determinant lemma after a rank- k update, gives the displayed determinant. $\square$

3. EXCLUSION OF THE COMPLETE EHLICH BLOCK FAMILY

The following finite lemma converts sign-Gram realizability into a condition on small block-sum sets.

Lemma 3.1 (Normalized block sums). *Let $r = (r_0, \dots, r_{s-1})$ be a partition of 23, set $G = E(r)$, and suppose that $G = HH^T$ for a sign matrix H . After independently changing column signs, every column q of H may be normalized to have first coordinate +1. If U is the block-incidence matrix and $z = U^T q$, then*

$$(6) \quad \frac{23}{20} + z^T C z = 1,$$

where

$$(7) \quad d_i = \frac{1}{4(5 + r_i)}, \quad \sigma = 1 - \sum_i r_i d_i, \quad C_{ij} = \frac{d_i d_j}{\sigma} - \frac{\delta_{ij}}{20(5 + r_i)}.$$

The finite domain is

$$\begin{aligned} z_0 &\in \{r_0, r_0 - 2, \dots, 2 - r_0\}, \\ z_i &\in \{r_i, r_i - 2, \dots, -r_i\} \quad (i > 0). \end{aligned}$$

Furthermore, if $z^{(1)}, \dots, z^{(23)}$ are the normalized column block sums, then

$$(8) \quad \sum_{j=1}^{23} z_a^{(j)} z_b^{(j)} = \begin{cases} 23r_a + 3r_a(r_a - 1), & a = b, \\ -r_a r_b, & a \neq b. \end{cases}$$

Proof. Changing column signs means replacing H by HD for a diagonal sign matrix D ; this preserves $HH^\top$ and conjugates $H^\top G^{-1}H = I$ to itself. The Sherman–Morrison formula applied to (4) gives

$$G^{-1} = \frac{1}{20}I + UCU^\top$$

with C as in (7). The diagonal entries of $H^\top G^{-1}H = I$ give (6). The normalized finite domains are simply the possible sums of signs in each block; in the first block one sign is fixed to $+1$. Finally, let s_a be the sum of the rows in block a . Then the left side of (8) is $\langle s_a, s_b \rangle$, and (4) evaluates this inner product. $\square$

For the sixteen candidates used below, put

$$\begin{aligned} A(z) &= z_0 z_1 - 1, & B(z) &= z_0 z_1 - z_0^2, \\ C_0(z) &= z_0 z_3 + z_1 z_2, \\ E_0(z) &= (z_2 + z_3)(10(z_0 + z_1) - z_2 + 5z_4), \end{aligned}$$

and

$$\begin{aligned} F(z) &= -75 + 25(z_0 z_3 + z_1 z_2) + 5(z_0 + z_1 + z_2 + z_3)(z_5 + z_6 + z_7) \\ &\quad + 8(z_5^2 + z_6^2 + z_7^2) - 9(z_5 z_6 + z_5 z_7 + z_6 z_7). \end{aligned}$$

Theorem 3.2 (Complete block-family exclusion). *No order-23 Ehlich block matrix $E(r)$ satisfying $\det E(r) > L^2$ is the Gram matrix of a sign matrix.*

Proof. There are 1,255 integer partitions of 23. Exact evaluation of (5) leaves 894 determinants above L^2 . In view of (3), only the sixteen square cases in Table 1 can be sign Grams.

For a row of the table, let $\mathcal{Z}(r)$ be the finite solution set of (6) in the domain of Lemma 3.1. Direct integer substitution shows that the indicated polynomial vanishes at every point of $\mathcal{Z}(r)$. On the other hand, summing that polynomial over the hypothetical 23 columns and applying (8) gives the nonzero integer in the last column. This contradiction excludes the candidate. When $\mathcal{Z}(r)$ is empty, even one normalized unit column is impossible. All cases are therefore excluded. $\square$

Remark 3.3. The table is a short certificate rather than a numerical optimization. The partition count, the determinant comparisons, every finite solution set of (6), every pointwise identity, and every moment sum are all recomputed with exact integers and rational numbers by the accompanying independent verifier.

4. A SIX-TOGGLE NEIGHBORHOOD OF THE BALANCED BLOCK FORM

Let A_0 be the adjacency matrix of $5K_4 \dot{\cup} K_3$ and put

$$G_0 = 24I - J + 4A_0 = E(4, 4, 4, 4, 4, 3).$$

TABLE 1. All square-determinant Ehlich block candidates above L^2 . The second column is $|\mathcal{Z}(r)|$ and the last is the prescribed sum of the identity over the 23 columns.

r	$ \mathcal{Z}(r) $	identity	sum
(1, 1, 1, 1, 1, 1, 3, 3, 11)	1	A	-24
(1, 1, 1, 3, 3, 3, 3, 11)	2	A	-24
(2, 2, 2, 8, 9)	4	B	-56
(1, 1, 1, 2, 2, 2, 5, 9)	2	A	-24
(2, 2, 5, 5, 9)	29	E_0	-1000
(1, 1, 1, 2, 2, 2, 2, 3, 9)	4	A	-24
(1, 1, 1, 1, 2, 2, 2, 4, 9)	4	C_0	-2
(1, 1, 1, 2, 6, 6, 6)	1	A	-24
(5, 5, 5, 8)	7	B	-200
(1, 1, 1, 2, 2, 2, 2, 2, 2, 6)	43	A	-24
(1, 1, 1, 5, 5, 5, 5)	37	A	-24
(1, 1, 1, 2, 2, 2, 2, 2, 5, 5)	0	-	-
(1, 1, 1, 2, 3, 5, 5, 5)	13	A	-24
(1, 1, 1, 1, 4, 5, 5, 5)	46	F	2800
(1, 1, 1, 4, 4, 4, 8)	3	A	-24
(2, 2, 2, 3, 3, 3, 3, 5)	3	B	-56

For an edge set $T \subseteq \binom{[23]}{2}$, let $A(T)$ be obtained by toggling the pairs in T , and define $G(T) = 24I - J + 4A(T)$. Thus every toggle switches one off-diagonal Gram entry between 3 and -1 .

Theorem 4.1 (Six-toggle exclusion). *If $|T| \leq 6$ and $\det G(T) > L^2$, then $G(T)$ is not the Gram matrix of an order-23 sign matrix.*

Proof. The automorphism group of the fixed partition is

$$\Gamma = (S_4^5 \rtimes S_5) \times S_3.$$

Canonical augmentation under this group gives the exact orbit counts in Table 2. The orbit-size sum in every layer equals $\binom{253}{t}$, so the list covers all labeled toggle sets. Determinants are evaluated by the matrix determinant lemma as fraction-free determinants of order at most 12 and checked on independent samples by direct 23×23 elimination.

By (3), only the square candidates in the last column can be sign Grams. Two of the five candidates through distance five are excluded by explicit local Hasse invariants, at $p = 37$ and $p = 2$, respectively. For every remaining square candidate, reduce $G^{-1} = Q/d$ with Q integral. If $G = HH^\top$ and the columns of H are normalized to start with $+1$, then they must be 23 distinct vertices of the graph

$$\mathcal{V}_G = \{q \in \{1\} \times \{\pm 1\}^{22} : q^\top Q q = d\}, \quad q \sim r \iff q^\top Q r = 0,$$

and must form a 23-clique, because $H^\top G^{-1} H = I$. Exact traversal of all 2^{22} normalized sign columns shows that each candidate graph is either empty or has an explicitly checked proper coloring with at most nine colors. Hence

none contains a 23-clique. This excludes all square cases and proves the theorem. $\square$

TABLE 2. Canonical toggle enumeration around G_0 . The orbit-size sum is an independent coverage check; the last column counts square determinants above L^2 .

t	orbit representatives	orbit-size sum	square cases
1	4	253	0
2	27	31,878	0
3	204	2,667,126	1
4	1,757	166,695,375	0
5	16,869	8,301,429,675	4
6	177,454	343,125,759,900	15

Remark 4.2. Theorem 4.1 is local in Gram-graph distance. No stability assumption placing an arbitrary high-determinant Gram matrix in this radius is used or claimed.

5. FIVE-ROW REPLACEMENT LOCAL OPTIMALITY

Fix the symmetric published record matrix H_0 with $|\det H_0| = L$. For five selected rows, write $H_0^{-1}e_{i_j} = c_j/s_j$ with integral c_j and positive integral s_j . Let $M = \text{lcm}(s_1, \dots, s_5)$ and form the 23 integer generators

$$w_t = ((M/s_1)c_1(t), \dots, (M/s_5)c_5(t)) \in \mathbb{Z}^5.$$

The quotient of all sign rows is the sign-sum zonotope

$$(9) \quad Z_S = \sum_{t=1}^{23} [-w_t, w_t].$$

By Lemma 2.1, the original five rows have determinant volume $T = M^5$ in these scaled coordinates. The implementation constructs the chamber arrangement from the integral forms c_j before applying the factors M/s_j . This loses no sign pattern: a positive diagonal coordinate change in the dual space gives a sign-preserving bijection between the two central arrangements.

Lemma 5.1 (Exact exterior reduction). *Let V be any finite set of sign sums in (9) containing all vertices. Put*

$$A_1 = \max_{x \in V} \|x\|^2, \quad A_2 = \max_{x, y \in V} \|x \wedge y\|^2.$$

If $A_1 A_2^2 \leq T^2$, no five replacement rows improve the determinant. Otherwise it is enough to retain bivectors p satisfying

$$(10) \quad \|p\|^2 A_2 A_1 > T^2$$

and pairs (p, r) satisfying

$$(11) \quad \|p\|^2 \|r\|^2 A_1 > T^2.$$

For every retained pair, the exact optimum over the fifth sign row is

$$(12) \quad \sum_{t=1}^{23} |n \cdot w_t|, \quad n = *(p \wedge r) \in \mathbb{Z}^5.$$

Proof. A multilinear function on Z_S^5 attains its maximum absolute value at vertices in each variable. For $p = x_1 \wedge x_2$ and $r = x_3 \wedge x_4$, the Hodge star is an isometry and Cauchy–Schwarz gives

$$|\det(x_1, \dots, x_5)|^2 \leq A_1 \|p\|^2 \|r\|^2.$$

This proves the coarse bound and both strict pruning rules. For a retained pair, multilinearity and independent choices of the 23 signs give

$$\max_{x_5 \in Z_S} |n \cdot x_5| = \sum_{t=1}^{23} |n \cdot w_t|,$$

which is (12). $\square$

Theorem 5.2 (Five-row local optimality). *The symmetric order-23 record matrix H_0 cannot be improved by arbitrary simultaneous replacement of at most five rows. The same is true for at most five columns.*

Proof. For each of the $\binom{23}{5} = 33,649$ row sets, the exact computation constructs a sign-point set containing all vertices of (9). Indeed, the central arrangement $w_t^\perp$ is essential. Every closed chamber is a pointed cone with an extreme ray contained in four independent arrangement hyperplanes. Enumerating all such four-tuples and all local zero-sign choices therefore includes every chamber sign pattern. Degenerate choices may add points, but every added point is still a genuine sign sum and is harmless.

Lemma 5.1 closes 22,173 row sets at the coarse bound. The remaining 11,476 are decided by (10)–(12), using 2,378,848,903 exact retained-pair support evaluations. In every case the maximum is exactly T , attained by the original rows; no value exceeds it. The complete result file contains each five-subset once; a separate validator reconstructs the targets and checks its aggregate integer fields. Independent verification of the refined cases is obtained by a fresh rebuild and rerun of the exact solver, not by treating the aggregate file as a trace of all support evaluations. Thus no five-row replacement improves $|\det H_0|$.

Any replacement of fewer than five rows is covered by adjoining unchanged rows until five indices have been selected. Since H_0 is symmetric, transposition converts a column replacement into the corresponding row replacement and proves the column statement. $\square$

Remark 5.3 (Arithmetic boundary). The arrangement and all products used in pruning comparisons are evaluated with arbitrary-precision integers.

The exterior stage uses fixed-width integers only after deriving input-specific bounds for every coordinate, Hodge product, and support sum; a row set is rejected rather than evaluated if a bound is not strictly inside the relevant signed range. Floating-point ratios and timings are diagnostic and are not used to accept a case.

6. REPRODUCIBILITY AND LIMITATIONS

The block-family theorem is verified from the statement alone by enumerating the integer partitions of 23, evaluating (5), deriving the inverse (7), and checking Table 1. The five-row theorem has separate stages for the integral inverse basis, the exhaustive row-set calculation, small algebraic unit tests, and a final completeness verifier. The discovery routines are not trusted by the final checks for the block theorem.

None of Theorems 3.2, 4.1, and 5.2 proves that $D(23) = L$. The block theorem treats all partitions in (4), but an arbitrary realizable Gram matrix need not have block form. The toggle theorem has a fixed radius and no global stability hypothesis. The replacement theorem rules out improvements at row distance at most five from one record representative, but does not control matrices farther away or classify all equality cases. A global solution still requires either a larger sign matrix or a complete, proof-producing classification of every realizable order-23 Gram matrix above L^2 .

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